



MARKERLESS SPORTS INTELLIGENCE

Motion Dynamics

AI-Powered Motion Intelligence for Sports and Human Performance For organizations committed to improving athletic performance and rehabilitation outcomes, our platform delivers actionable insights into movement, efficiency, and recovery—helping businesses enhance results, engagement, and client success.

Start motion



SERVICE 01

Scalable Data Intelligence

Train models using match footage and open datasets to build reliable tactical, biomechanical, and movement data pipelines.

SERVICE 02

Real-Time Performance Feedback

Deliver live insights to coaches, analysts, and athletes during training and competition, improving decision-making in real time.

SERVICE 03

Automated Match & Player Insights

Extract frame-level data, analyze performance patterns, and generate tailored reports, recommendations, and predictive match outcomes.



PRECISION SPORTS ANALYTICS

Multi Sport applications. Markerless Tracking. 95% Accuracy.

Delivering simple, scalable AI-powered motion capture and biomechanical analytics that integrate seamlessly into your workflow—driving measurable ROI for sports teams, training centers, and institutions worldwide.

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Built for impact

We combine computer vision and large language models to transform raw movement into insight, coaching, and measurable impact.

SOLUTION

Automated Match & Player Insights

Extract frame-level data, analyze performance patterns, and generate tailored reports, recommendations, and predictive match outcomes.

SOLUTION

AI Coaching & Insights (LLMs)

Large language models convert biomechanics into natural feedback: session summaries, player-specific reports, and goal-led plans coaches and athletes can act on instantly.

SOLUTION

Rehabilitation & Research (B2B)

Secure data pipelines, custom dashboards, and integrations with hospitals and universities. License datasets, evaluate interventions, and accelerate research with explainable metrics.



Industries We Serve

Purpose-built for organizations across sport, media, and healthcare.

INDUSTRY

Sports Clubs

Player monitoring, skills tracking, and talent ID.

INDUSTRY

Broadcasters

Real-time graphics, highlights, and automated storytelling.

INDUSTRY

Rehab Clinics

Progress tracking and personalized recovery insights.

INDUSTRY

Universities

Research integrations, data access, and validation.



TESTIMONIALS

Teams turning movement into measurable outcomes

Stories from sport, media, rehab, and research teams using Motion Dynamics to make performance intelligence easier to capture, share, and act on.



ACADEMY PERFORMANCE

3x

How a youth performance lab turned weekly training video into coaching-ready movement reports.

Motion Dynamics helped our coaches move from subjective review to clear, repeatable evidence after every session.

Performance Director

Junior Performance Lab

FASTER REVIEW CYCLES



BROADCAST ANALYTICS

Live

How a media team transformed raw match footage into real-time tactical graphics.

The automated overlays made complex movement stories simple enough for fans to understand instantly.

Production Lead

Sports Media Studio

GRAPHICS PIPELINE



REHAB TRACKING

95%

How clinicians used markerless tracking to monitor recovery without slowing appointments.

We can show athletes progress in plain language while still keeping the data rigorous enough for clinicians.

Clinical Lead

Movement Rehab Clinic

TRACKING ACCURACY TARGET



UNIVERSITY RESEARCH

Multi

How researchers connected motion capture, AI models, and repeatable validation workflows.

The platform gave our team a practical bridge between lab-grade analysis and real-world sport footage.

Research Partner

University Sports Lab

SPORT DEPLOYMENT



CLUB OPERATIONS

0

How a club identified technical trends across players without adding wearable sensors.

Coaches finally had a shared language for movement quality, workload, and development priorities.

Head Coach

Elite Training Club

WEARABLES REQUIRED



TALENT ID

Frame



How scouting teams compared athletes using consistent biomechanical and tactical indicators.

Instead of isolated clips, we now review structured signals that make selection meetings more objective.

Talent Lead

Performance Pathway Team

LEVEL INSIGHTS

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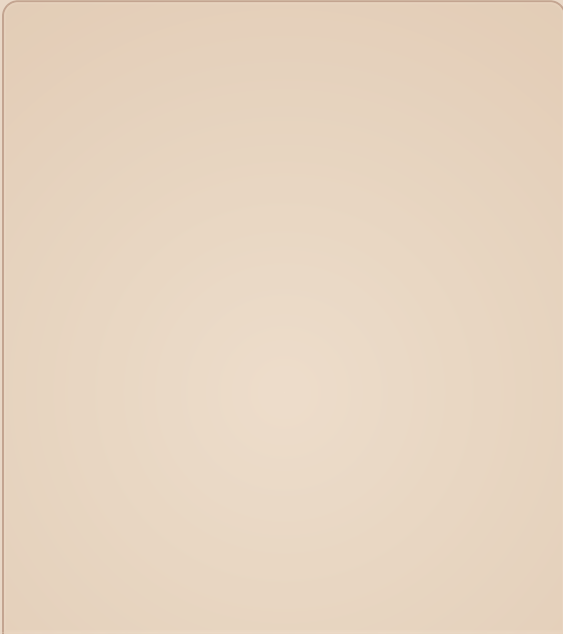
DEMO

Interactive swing intelligence demo

Explore the same movement metrics coaches, analysts, and athletes use to understand speed, launch, path, face control, and shot outcome.

3D MODEL

The mannequin



Club Speed

88 mph

Ball Speed

101 mph

Carry Distance

114 yd

Total Distance

127 yd

Smash Factor

1.9 deg

Attack Angle

2.1 deg

Launch Angle

2.3 deg

Dynamic Loft

2.5 deg

Spin Axis

3 deg

SELECTED METRIC

Club Speed

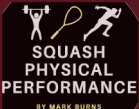


Club Path	4 deg
Face Angle	5 deg
Face to Path	6 deg
Shot Dispersion	7 deg
Shot Apex	8 ft

COLLABORATIONS

Collaborations & Partners

We team up with leading organizations across sport, media, and research.



TEAM

Built by engineers, athletes, and researchers.

An industrial-grade team focused on shipping real-world motion intelligence, fast, reliable, and measurable.

02

Core team

Diar
Abdlkarim

Designs brain-informed technologies that enhance perception, movement, and human-machine collaboration through HCI, VR, BC, and AI-based motion analytics.

Diar Abdlkarim

POST-DOCTORAL RESEARCH FELLOW
Founder and leader

Stuart
MacGregor

Former professional athlete with sports data engineering and research experience across artificial intelligence, motion capture, and enhanced sports experiences.

Stuart MacGregor

MSC COMPUTER SCIENCE GRADUATE
Founder and leader



Business Supports

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Max Di Luca

Researches how the human brain processes multidimensional information for perception and action using psychophysical methods and computational models.

Max Di Luca

ASSOCIATE PROFESSOR

Co-founder

John Cook

Business Advisor

John Cook

BUSINESS ADVISOR

Max Ward

Business Advisor

Max Ward

BUSINESS ADVISOR

05

Technical Team

Vishwanath Nair

Machine Learning Engineer

Vishwanath Nair

MACHINE LEARNING ENGINEER

Angel Yao

Machine Learning Engineer

Angel Yao

MACHINE LEARNING ENGINEER

Genia Penksik

Co-Founder

Genia Penksik

TECHNICAL CONSULTANT

Co-Founder

Melika Hashemi

Full Stack Developer

Melika Hashemi

FULL STACK DEVELOPER

Erfan Echresh

Full Stack Developer

Erfan Echresh

FULL STACK DEVELOPER

01

Marketing Team

Maria Kononova

Marketing

Maria Kononova

MARKETING



Jonathan Tate
UNIVERSITY OF BIRMINGHAM
HEAD SQUASH COACH

CONTACT

Get In Touch

Ready to accelerate your sports development? Contact our team today.

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EMAIL

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YOUR MESSAGE

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The future of sport development through advanced motion capture and biomechanical precision.

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